

Hannah Wichern

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About me

PhD student in astronomy with a special interest in rare transients, their multi-wavelength emission, and environments. I am a curious person who enjoys being challenged, and who has experience with observing, modelling, and analysing data. I value being able to work independently while also being able to share ideas in an open and creative atmosphere.

Education

PhD in Transient Astronomy

Nov. 2023 - Oct. 2026

Technical University of Denmark - Kongens Lyngby, Denmark

Advisor: Giorgos Leloudas, co-advisors: Mattia Bulla, Jérôme Chenevez

MSc in Physics and Astronomy (Cum Laude)

Sept. 2021 - Aug. 2023

Radboud University - Nijmegen, Netherlands

Advisors: Maria Ravasio, Peter Jonker

BSc in Physics and Astronomy

Sept. 2018 - June 2022

Radboud University - Nijmegen, Netherlands

Publications (as first author)

My journal publications can be found on ORCID: <https://orcid.org/0009-0004-1442-619X>.

- H. C. I. Wichern et al. (in preparation), *A superluminous supernova in a quiescent galaxy*.
- H. C. I. Wichern et al. (in preparation), *Reducing Monte Carlo noise in radiative transfer simulations with a trajectory-based technique*.
- H. C. I. Wichern, G. Leloudas, M. Pursiainen, et al. (2026). *Revisiting the unification of tidal disruption events with polarimetry*. *Astronomy & Astrophysics*, 705, A250.
- H. C. I. Wichern, M. E. Ravasio, P. G. Jonker, et al. (2024). *Investigating the off-axis GRB afterglow scenario for extragalactic fast X-ray transients*. *Astronomy & Astrophysics*, 690, A101.

Grants and awards

- AHEAD 2020 Trans-national access in computational astrophysics, call A07: *The 3D geometry of transients related to gravitational wave events* (travel grant)
- KHMW (Royal Holland Society of Sciences and Humanities) Young Talent Award 2023 graduation prize in astrophysics.

Awarded time (as PI)

- Chandra X-ray Observatory DDT proposal ObsID 32275 (15 ks with ACIS-S)
- Very Large Telescope P117 proposal 117.29QM.001 (1.3 hours with MUSE)
- Danish e-infrastructure Consortium (DeiC) National HPC resources DeiC-DTU-N5-2026180 (850k CPU hours on the LUMI supercomputer)
- Nordic Optical Telescope (NOT) fast-track proposal 71-418 (3 hours with ALFOSC)
- NOT/IDA Young Researcher Call proposal 69-854 (8 hours with ALFOSC, visitor mode)

Teaching activities

Technical University of Denmark Co-supervision Bachelor research project.	Feb. - June. 2026
Technical University of Denmark Teaching assistant for Bachelor course <i>Geo- and Planetary Physics I</i> .	Sept. - Dec. 2024, 2025
Radboud University Teaching assistant for Bachelor course <i>Space Astronomy</i> .	Apr. - June 2023
Radboud University Teaching assistant for Bachelor course <i>Extreme Astrophysics</i> .	Nov. 2022 - Jan. 2023

Conferences and workshops

European Astronomical Society Annual Meeting (Lausanne, Switzerland) Contributed talk: <i>A massive and passive galaxy harbouring a superluminous supernova.</i>	June 2026
IAUS 406 – Future landscape of astrophysical transients (Turku, Finland) Contributed talk: <i>A curious superluminous supernova in a passive galaxy.</i>	May 2026
European Astronomical Society Annual Meeting (Cork, Ireland) Contributed talk: <i>All shapes and sizes? Unification of polarised tidal disruption events.</i>	July 2025
X-ray Quasi-Periodic Eruptions and Repeating Nuclear Transients (Madrid, Spain) Contributed talk: <i>Peering into the photospheres of tidal disruption events.</i>	June 2025
Tidal Disruption Events and Nuclear Transients: Entering the Data-Rich Era (Heraklion, Greece) Contributed talk: <i>Shedding light on the photosphere evolution of tidal disruption events with polarimetry.</i>	Sept. 2024
European Astronomical Society Annual Meeting (Padova, Italy) Contributed talk: <i>Investigating the off-axis GRB afterglow scenario for extragalactic fast X-ray transients.</i>	July 2024
Annual Danish Astronomy Meeting (Roskilde, Denmark) Poster: <i>Off-axis GRB afterglows as an origin for extragalactic fast X-ray transients.</i>	May 2024
ESO La Silla Observing School (Santiago de Chile, Chile)	Feb. 2024

Memberships

- Rubin-ESO Investigative Spectroscopy Of Transient Targets and Objects (RISOTTO) collaboration (Feb. 2026 - present)
- Vera C. Rubin Observatory LSST Transients and Variable Stars collaboration (Dec. 2025 - present)
- BlackGEM collaboration (Nov. 2023 - present)
- Public ESO Spectroscopic Survey of Transient Objects (PESSTO) collaboration (Nov. 2023 - present)
- Electromagnetic counterparts of gravitational wave sources at the Very Large Telescope (ENGRAVE) collaboration (Nov. 2023 - present)